



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 2 – Overloading and Inheritance
Topic – 2.4 Introduction to Inheritance

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Introduction

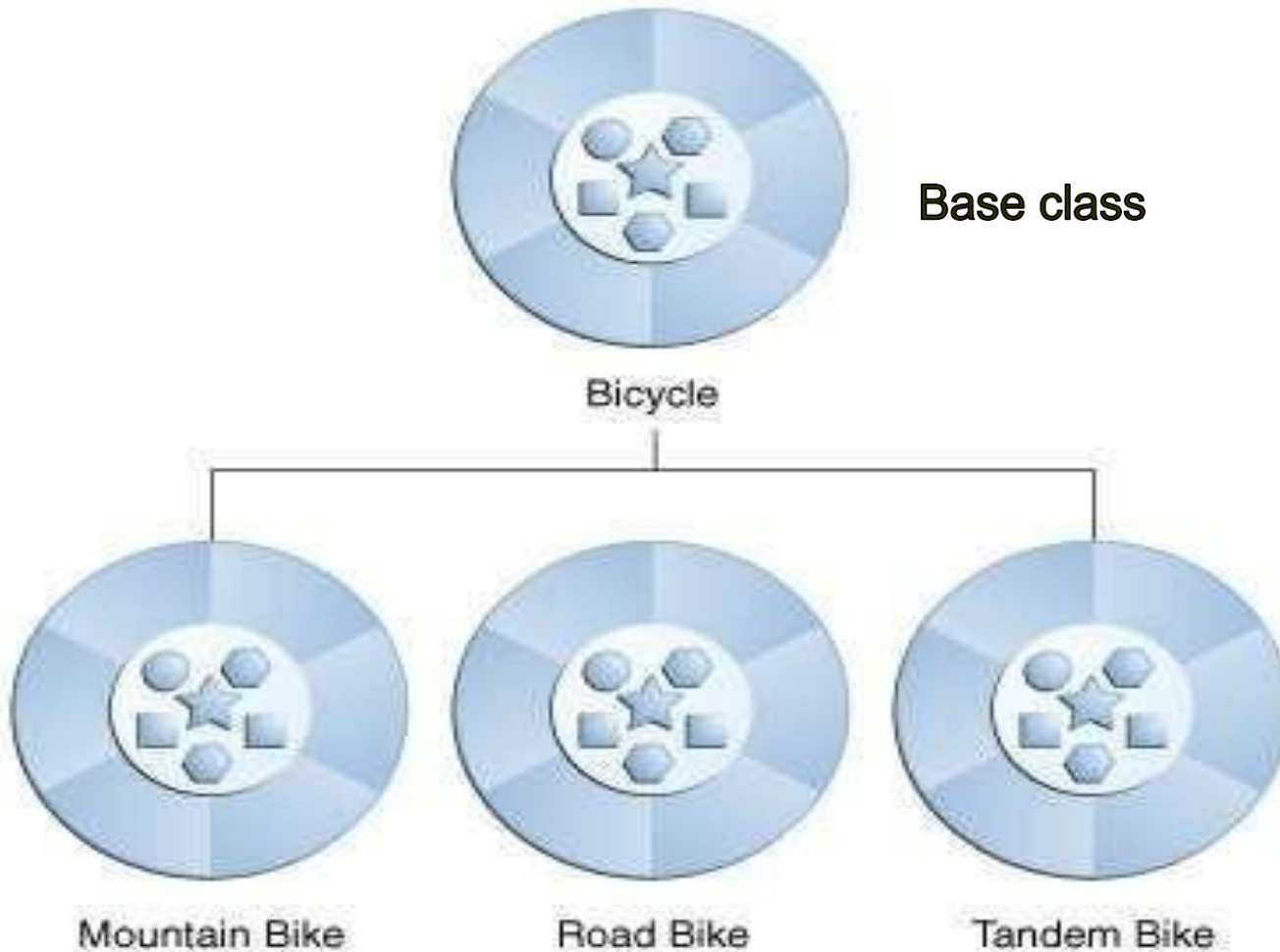
- Inheritance in C++ is one of the major aspects of Object Oriented Programming.
- It is the process by which one object can inherit or acquire the features of another object.
- It is a way of creating new class(derived class) from the existing class(base class) providing the concept of reusability.



Introduction

- The base class is called as superclass and derived class is called a subclass.
- Semantically, inheritance denotes an “*is-a*” relationship between a class and one or more refined version of it.

Example



Base class

“Bicycle” is a **generalization** of “Mountain Bike”.

“Mountain Bike” is a **specialization** of “Bicycle”.

Derived
Classes



Defining a Base Class

- Base class has general features which are common to all the derived classes and derived class adds specific features.
- This enables us to form a hierarchy of classes.

```
class Base-class
```

```
{
```

```
... ..
```

```
.....//Members of base class
```

```
};
```



Defining a Derived Class

- The general form of deriving a subclass from a base class is as follows:

```
Class derived-class-name : visibility-mode base-class-name
{
    .....
    .....// members of the derived class
};
```

- The visibility-mode is optional, it may be either private or public or protected, by default it is private.
- This visibility mode specifies how the features of base class are visible within the derived class.



Example using public inheritance

```
#include <iostream>
using namespace std;
class Animal
{
public:
    void eat()
    {
        cout<<"Eating..."<<endl;
    }
};
class Dog: public Animal
{
public:
```

```
    void bark()
    {
        cout<<"Barking...";
    }
};
int main()
{
    Dog d1;
    d1.eat();
    d1.bark();
    return 0;
}
```

Output:

Eating...

Barking...



Example using private inheritance

```
#include <iostream>
using namespace std;
class A
{
    int a = 4;
    int b = 5;
public:
    int mul()
    {
        int c = a*b;
        return c;
    }
};

class B : private A
{
public:
```

```
    void display()
    {
        int result = mul();
        cout <<"Multiplication of a and b is : "<<result;
    }
};

int main()
{
    B b;
    b.display();
    return 0;
}

Output:
Multiplication of a and b is : 20
```




Thank You..